

AQUINAS DATASET HANDBOOK

INTRODUCTION

The **AQUINAS** Dataset (**A**vailable **Q**uantities **I**ntended for **A**nalysis and **S**cience) is a set of measurements recorded over a few years on a real bridge, made available for research purposes by OSMOS Group. The measurements include both acceleration (24 sensors) and strain (24 sensors) on different locations on the bridge deck at a sampling rate of 100 Hz. The recording is trigger-based: the dataset is split into individual records, each one with a duration of a few seconds, corresponding to the passage of vehicles on the bridge, and listed in index tables.

DESCRIPTION OF THE BRIDGE AND SENSOR LOCATIONS

The bridge remains anonymous in the frame of research activities using the AQUINAS dataset. It is a double prestressed concrete deck with 40 to 50 m long spans: the upstream deck and the downstream deck are independent, and the records do not happen at the same time for both, since each one of them carries the road in one single direction. The two decks are very similar in terms of geometry, they also have the same span lengths, although some details of the cross-sections are different: for example, concrete thickness and prestress cables. The upstream deck was built more than 10 years before the downstream deck.

Sensors are set on the two first spans of each deck, on six different cross-sections:

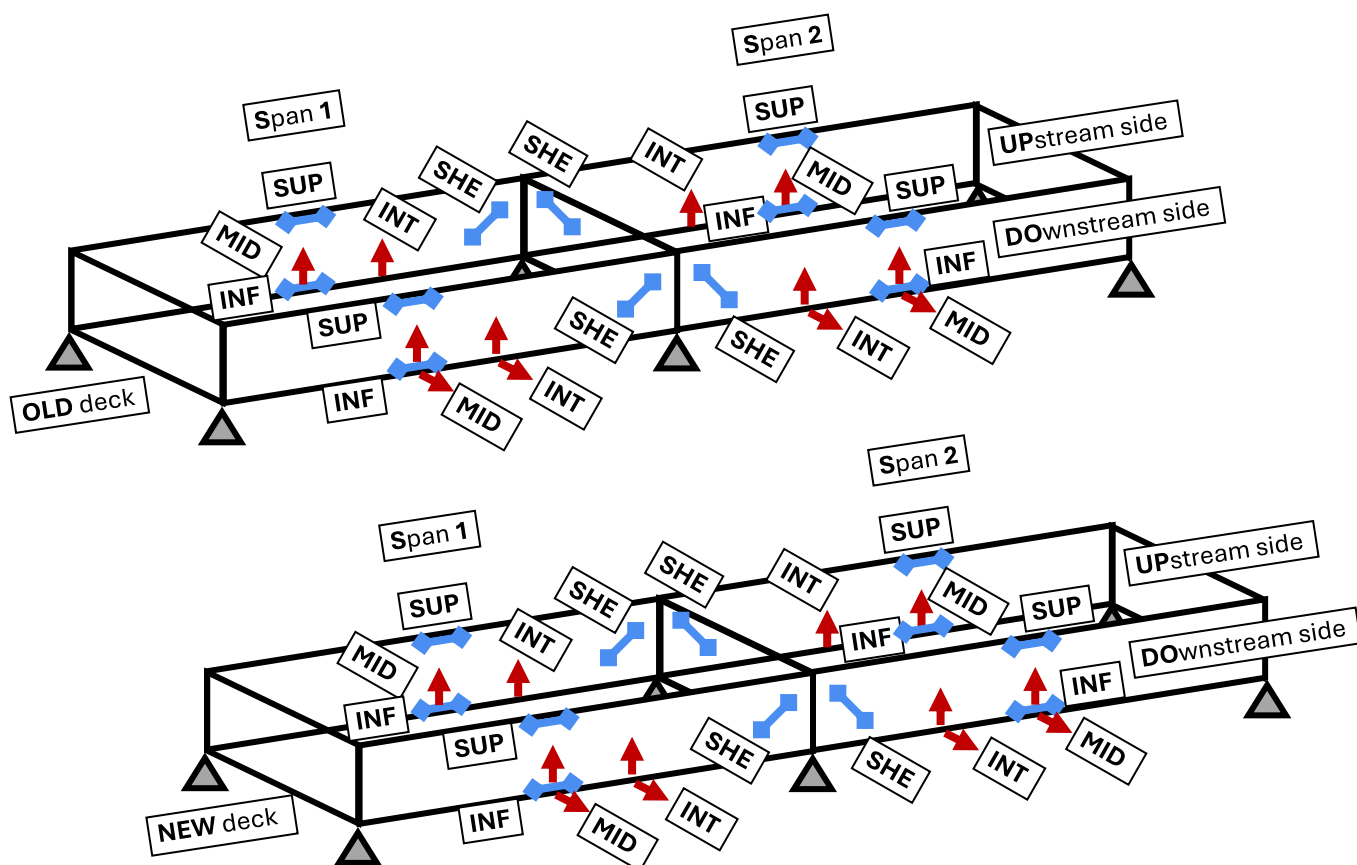
Section name	Location	Acceleration sensors	Strain sensors
S1_MID	½ length of span 1	2x Z-axis + 1x Y-axis	4x Longitudinal
S1_INT	2/3 length of span 1	2x Z-axis + 1x Y-axis	None
S1_SHE	Near pier 1 on span 1	None	2x Diagonal (SHE ar effect)
S2_SHE	Near pier 1 on span 2	None	2x Diagonal (SHE ar effect)
S2_INT	1/3 length of span 2	2x Z-axis + 1x Y-axis	None
S2_MID	½ length of span 2	2x Z-axis + 1x Y-axis	4x Longitudinal

The naming of each sensor gives all indications on its location:

Information	Deck	Span	Side	Location	Quantity	Axis direction
Possible values	OLD (old upstream deck)	S1 (span 1)	UP (upstream)	MID (mid-span) INT (intermediate = 1/3 of span length)	ACC (acceleration in g = 9.81 m/s ²)	Y (transversal) Z (vertical)
	NEW (new downstream deck)	S2 (span 2)	DO (downstream)	INF (inferior fibre) SUP (superior fibre) SHE (on the web, set in diagonal for shear strain)	STR (strain ε in ‰ = 1 mm/m)	(not relevant for strain sensors)

SCHEMATIC VIEW OF THE SENSOR LOCATIONS

-  **STRain sensor**
-  **ACCelerometer in direction Y**
-  **ACCelerometer in direction Z**



STRUCTURE OF THE DATASET

The AQUINAS dataset V.1.0 is made of 5 datasets, each one for one full calendar month: July 2022, April 2023, August 2023, January 2024, and June 2024.

For each month, the dataset includes:

- 48 index tables in JSON format, one for each sensor,
- 48 directories, one for each sensor, with raw data files in JSON format, split by calendar days.

The index tables give 15 basic features for each record, as follows:

Feature name in JSON file	Description	Example
Record_UID	Unique ID number of the record*	16341
File	Name of the JSON file with the raw data of the record, to be found in the directory named after the sensor	"OLD_S2_UP_SUP_STR_SET1_1.json"
Start_Row	First row of the record in the raw data JSON	1
End_Row	Last row of the record in the raw data JSON	1527
Start_Time	UTC timestamp of the start of the record	"2022-07-01 01:47:16"
End_Time	UTC timestamp of the end of the record	"2022-07-01 01:47:31"
Duration	Duration of the record in seconds	15.26
Start_Value	First value of the record (in g or ‰)	-0.1184
End_Value	Last value of the record (in g or ‰)	-0.1187
Diff_Value	End_Value minus Start_Value (in g or ‰)	-0.0002
Min_Value	Minimum value of the record (in g or ‰)	-0.1416
Max_Value	Maximum value of the record (in g or ‰)	-0.116
Mean_Value	Mean value of the record (in g or ‰)	-0.1197
Range	Max_Value minus Min_Value (in g or ‰)	0.0256
Temperature	Ambient temperature measured at the time of the record in °C	13.3159

***Record_UID** is specific for each sensor and doesn't match with other sensors. To carry out a record matching between different sensors, please use **Start_Time** and **End_Time** features instead.

Note that rounding artifacts may affect the values of the features **Diff_Value** and **Range**.

For example: to extract the raw data of the strain sensor **OLD_S2_UP_SUP_STR** for the record number **16341**, starting **2022-07-01 01:47:16** and ending **2022-07-01 01:47:31**:

Open the **OLD_S2_UP_SUP_STR** directory, then open the file **OLD_S2_UP_SUP_STR_SET1_1.json**, and read the rows **1** to **1527** for both columns: **timestamp** (in UTC) and **OLD_S2_UP_SUP_STR** (measurement values of the sensor).

This organization of the dataset in multiple small JSON files is intended to reduce the access time to raw data, in the case of a study for a short period on few sensors only. It allows also to flexibly share only parts of the dataset, by selecting the files related to a reduced number of sensors and shorter time periods.